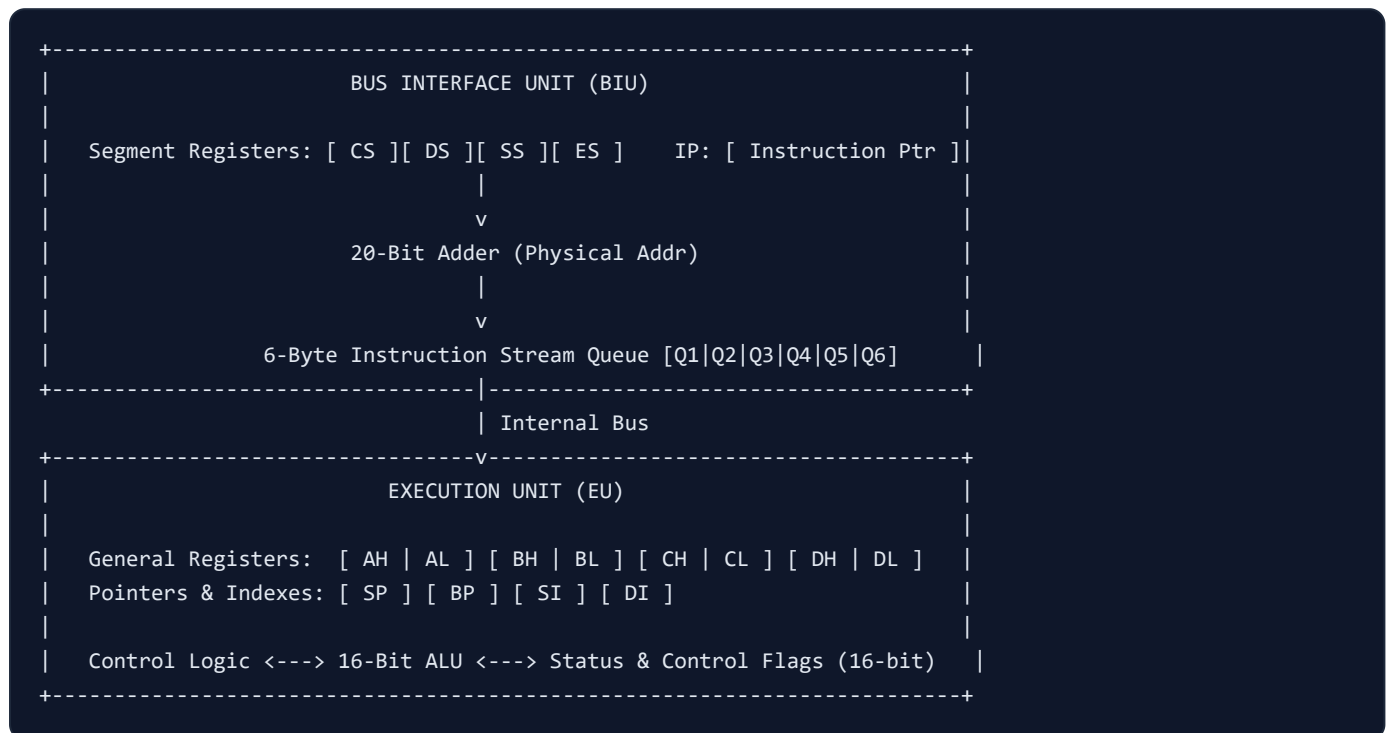


5-Mark Questions & Answers — MPCA Unit 3: The Intel Microprocessor 8086 Architecture

Q1. Explain the internal architecture of 8086 (BIU and EU) with a neat diagram.



- 1. Bus Interface Unit (BIU):** Handles external bus communication, fetches instruction code into a 6-byte FIFO queue, and computes 20-bit physical addresses using segment registers (CS, DS, SS, ES) and IP.
- 2. Execution Unit (EU):** Receives prefetched opcodes from queue, decodes micro-instructions, executes operations using the 16-bit ALU, and updates general registers and 9 active flag bits.

Q2. Explain the programmer's model of 8086, listing all registers and their functions.

- 1. Data Registers (16-bit / 8-bit split):**
 - **AX** (AH/AL): Accumulator (used in multiplication, division, I/O).
 - **BX** (BH/BL): Base register (holds memory base address).
 - **CX** (CH/CL): Count register (loop counter, shift count).
 - **DX** (DH/DL): Data register (I/O port addressing, high-word math).
- 2. Pointer & Index Registers:**
 - **SP** (Stack Pointer), **BP** (Base Pointer), **SI** (Source Index), **DI** (Destination Index).
- 3. Segment Registers:**
 - **CS** (Code Segment), **DS** (Data Segment), **SS** (Stack Segment), **ES** (Extra Segment).
- 4. Instruction Pointer & Flags:**
 - **IP** (Instruction Pointer), **FLAGS** (9 active flags: CF, PF, AF, ZF, SF, TF, IF, DF, OF).

Q3. Draw and explain the functional pin diagram of 8086 in detail.

- **40-Pin DIP Package Configuration:**
- **AD0 - AD15** (Pins 2-16, 39): Multiplexed Address/Data Bus.
- **A16/S3 - A19/S6** (Pins 35-38): Multiplexed Address/Status Bus.
- **BHE' /S7** (Pin 34): Bus High Enable / Status S7.
- **RD'** (Pin 32): Read control output.
- **READY** (Pin 22): Wait state insertion signal from memory.
- **INTR** & **NMI** (Pins 17, 18): Interrupt request inputs.
- **RESET** (Pin 21): System reset input.
- **MN/MX'** (Pin 33): Minimum / Maximum mode select pin.

Q4. Explain memory segmentation and physical address calculation in 8086 with an example.

Physical Address Formula

Physical Address = (Segment Register Value × 16) + Offset Register Value

Numerical Example

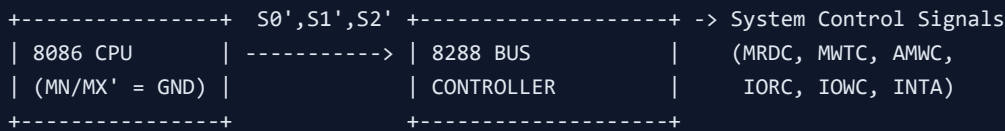
- **Given:** Code Segment **CS = 2500H**, Instruction Pointer **IP = 0124H**.
- **Calculation:**
CS Value in Hex = 2500H Shifted Segment Base (CS × 16) = 25000H Physical Address = 25000H + 0124H = 25124H
- **Result:** The 20-bit address generated on the physical address bus is **25124H**.

Q5. Explain the functioning of 8086 in minimum mode with a diagram.

```
+-----+ ALE +-----+ -> A0 - A19 Address Bus
| 8086 CPU | ----> | Latches (74LS373) |
| (MN/MX' = +5V) | +-----+
|           | AD0-AD15 +-----+ -> D0 - D15 Data Bus
|           | -----> | Transceivers |
|           |           +-----+
|           | Control +-----+
|           | -----> | Memory / I/O |
+-----+ (RD,WR) +-----+
```

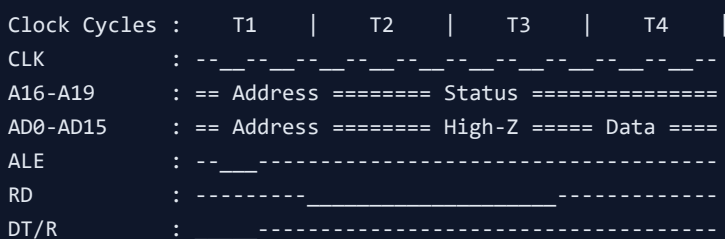
- In Minimum Mode (**MN/MX' = 1**), the 8086 operates as a standalone processor.
- Control signals (**RD**, **WR**, **M/IO**, **ALE**, **DT/R**, **DEN**) are generated directly by the 8086 CPU pins without external bus controllers.

Q6. Explain the functioning of 8086 in maximum mode with a diagram.



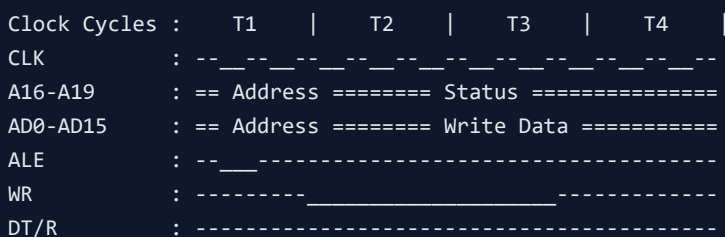
- In Maximum Mode ($MN/MX' = 0$), status outputs $S0'$, $S1'$, $S2'$ are sent to an external **Intel 8288 Bus Controller**.
- The 8288 decodes these status lines to generate system memory and I/O command control signals, allowing multi-processor or co-processor (8087 NDP) operation.

Q7. Draw and explain the timing diagram for a read operation in minimum mode.



- **T1:** **ALE** goes HIGH; 20-bit address output on **AD0-AD15** and **A16-A19** .
- **T2:** **ALE** goes LOW; **AD0-AD15** enters High-Z mode; **RD** signal drops LOW.
- **T3:** Memory places valid data on data bus **AD0-AD15** .
- **T4:** **RD** returns HIGH; CPU latches data into destination register.

Q8. Draw and explain the timing diagram for a write operation in minimum mode.



- **T1:** **ALE** goes HIGH; target address output on **AD0-AD15** .
- **T2:** CPU outputs valid write data onto **AD0-AD15** and drops **WR** line LOW.
- **T3:** Data remains stable on data bus; memory writes data into RAM address.
- **T4:** **WR** returns HIGH; bus returns to idle state.

Q9. Explain interrupt structure and interrupt servicing in 8086.

1. Interrupt Types:

- **Hardware Interrupts:** External pin signals — Non-Maskable Interrupt (**NMI** , Type 2) and Maskable Interrupt

(**INTR**).

- **Software Interrupts:** Triggered by instructions — **INT n** (Type 0-255), **INTO** (Overflow), **INT 3** (Breakpoint).

2. Interrupt Vector Table (IVT):

- Located at memory range **00000H** to **003FFH** (1KB size).
- Contains 256 vector pointers (4 bytes per vector: 2 bytes **IP** + 2 bytes **CS**).

3. Servicing Sequence:

- Push **FLAGS** , clear **IF** and **TF** , push **CS** , push **IP** .
- Read vector address **4 × Type Number** , load **CS:IP** , and jump to ISR.